

INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH MOHALI

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Summer Research Internship Report

Classification and Decomposition of Alternating Tensors

Theory and Computational verification in Low-Dimensional Exterior Powers

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Indian Institute of Science Education and Research Mohali
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Certificate

This is to certify that the Summer Research Internship Report entitled **"Classification and Decomposition of Alternating Tensors: Theory and Computational verification in Low-Dimensional Exterior Powers"** submitted by **Akarsh Suryavanshi**, M.Sc. Mathematics, National Institute of Technology Durgapur, is a bonafide record of the research work carried out by him under my supervision during the Summer Research Internship at the Department of Mathematical Sciences, Indian Institute of Science Education and Research Mohali, during May–July 2026.

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Abstract

We study the decomposability and orbit classification of alternating tensors under the action of the general linear group. Motivated by the geometry of Grassmannians and the Plücker embedding, we investigate the decomposition of alternating tensors in $\Lambda^2(\mathbb{R}^4)$ and $\Lambda^3(\mathbb{R}^6)$. In case of $\Lambda^2(\mathbb{R}^4)$, we establish the decomposability criterion $\omega \wedge \omega = 0$ and recover the orbit classification of bivectors. For $\Lambda^3(\mathbb{R}^6)$, we implement Bryant's classification computationally using the contraction map and Hitchin's quartic invariant, providing an algorithm that classifies arbitrary alternating 3-forms into their corresponding $GL(6, \mathbb{R})$-orbits. We verify computationally the invariance of contraction rank under change of basis and illustrate the Hitchin decomposition of stable forms.we perform exploratory computational study of $\Lambda^3(\mathbb{R}^7)$, where extensive random sampling suggests that contraction rank 4 may not occur, although no theoretical proof is provided.

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Chapter 1

Introduction

1.1 Motivation

Geometric quantities such as oriented area and oriented volume can be represented by alternating tensors. This motivates the study of decomposition problems in exterior powers, where the objective is to express an alternating tensor as a sum of the smallest possible number of decomposable tensors.

Since decomposability is an intrinsic property of a tensor, it is independent of the choice of basis. Consequently, it is natural to study tensors up to the action of the general linear group $\mathrm{GL}(V)$, which preserves both decomposability and the number of decomposable summands. This reduces the problem to understanding the orbit structure of $\Lambda^r(V)$ under $\mathrm{GL}(V)$ and identifying simple representatives for each orbit.

Dimension-count arguments further indicate that the decomposition problem becomes increasingly complex near the middle exterior powers. As shown in Table 1.1, the expected lower bound on the number of decomposable summands attains larger values when r is close to $\frac{\dim(V)}{2}$.

Motivated by these observations, this report develops both theoretical and computational approaches for the cases $\Lambda^2(\mathbb{R}^4)$ and $\Lambda^3(\mathbb{R}^6)$.

1.2 Problem Statement

Given an n-dimensional vector space V and an integer $1 \leq r \leq n$,we need to determine the smallest integer $m = m(r, n)$ such that every element of $\Lambda^r(V)$ can be expressed as a sum of at most m decomposable alternating r-vectors.

1.3 Objectives

- Derive an expected lower bound on $m(r, n)$ from the dimension of secant varieties of the Grassmannian $G(r, \mathbb{R}^n)$.
- Establish necessary and sufficient conditions for decomposability of a 2-vector in $\Lambda^2(\mathbb{R}^4)$ and classify $\Lambda^2(\mathbb{R}^4)$ into $GL_4(\mathbb{R})$-orbits.
- Implement Plücker relations computationally to test decomposability of 3-vectors in $\Lambda^3(\mathbb{R}^6)$, and identify $GL(6, \mathbb{R})$-invariants suitable for orbit classification.

- Implement Bryant's classification of $\Lambda^3(\mathbb{R}^6)$ into six normal forms computationally using the contraction rank and Hitchin's quartic invariant.
- Extend the computational framework to $\Lambda^3(\mathbb{R}^7)$ as an exploratory study of contraction ranks beyond the classified case.

1.4 Geometric Interpretation of the Problem

This section gives the geometric viewpoint behind the decomposition problem.

1.4.1 Problem Setup

Consider m be the smallest integer such that every element of the r-th exterior power $\Lambda^r(\mathbb{R}^n)$ is expressible as a sum of m decomposable tensors.

Consider the Grassmannian $G(r, \mathbb{R}^n)$ consisting of all r-dimensional linear subspaces of $\mathbb{R}^n$.

The Plücker embedding is given by

$$G(r, \mathbb{R}^n) \hookrightarrow \mathbb{P}(\Lambda^r \mathbb{R}^n)\,, \qquad \langle v_1, v_2, \ldots, v_r \rangle \longmapsto [v_1 \wedge v_2 \wedge \cdots \wedge v_r].$$

What sum of m decomposable look like.

Pick one point from $G(r, \mathbb{R}^n)$,It represent one decomposable tensor..

Pick two points from $G(r, \mathbb{R}^n)$, say $P_1, P_2$and add them linearly :

$$\alpha P_1 + \beta P_2,$$

it represent the line joining P_1 and P_2.

Take the space of all lines joining every pair of points of $G(r, \mathbb{R}^n)$,

$$\sigma_2(G(r, \mathbb{R}^n))\,.$$

It represents every tensor that can be written as the sum of at most two decomposable tensors.

Hence our question boils down to :

How many points of $G(r, \mathbb{R}^n)$ are required so that their span covers $\mathbb{P}(\Lambda^r \mathbb{R}^n)$?

1.4.2 Expected Dimension of Secant Varieties

[Lan12]

Dimension of $G(r, \mathbb{R}^n)$

For defining an r-dimensional linear subspace we need r independent vectors.

Each vector has n parameters, so we begin with rn parameters.

However, different choices of basis may represent the same subspace.

If the basis vectors are arranged as an $r \times n$ matrix,

$$\begin{bmatrix} e_{11} & e_{12} & \cdots & e_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ e_{r1} & e_{r2} & \cdots & e_{rn} \end{bmatrix},$$

then multiplying on the left by any invertible $r \times r$ matrix changes only the basis and not the subspace.

Therefore,

$$\dim G(r, \mathbb{R}^n) = nr - r^2 = r(n-r).$$

An example is shown below.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 & 10 \\ 0 & 0 & 9 \end{bmatrix} = \begin{bmatrix} 2 & 0 & 28 \\ 6 & 0 & 66 \end{bmatrix} \xrightarrow{\text{row operations}} \begin{bmatrix} 2 & 0 & 28 \\ 0 & 0 & -18 \end{bmatrix} \xrightarrow{\text{row operations}} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 9 \end{bmatrix}.$$

Expected Dimension

On each point of $G(r, \mathbb{R}^n)$ there are $nr - r^2$ parameters.

Choosing m points gives $m(nr - r^2)$ parameters.

Taking all linear combinations of these m points contributes another $m-1$ parameters.

Hence the expected dimension is

$$m(nr - r^2) + m - 1.$$

To cover the whole projective space $\mathbb{P}(\Lambda^r \mathbb{R}^n)$, we require

$$m(nr - r^2) + m - 1 \geq \binom{n}{r} - 1.$$

Therefore,

$$m \geq \frac{\binom{n}{r}}{r(n-r)+1}.$$

Since m is an integer,

$$m = \left\lceil \frac{\binom{n}{r}}{r(n-r)+1} \right\rceil.$$

Table 1.1: Expected lower bound for the minimum number of decomposable tensors obtained from the dimension count.

(a) $2 \leq n \leq 6$

n	r	$\left\lceil \frac{\binom{n}{r}}{r(n-r)+1} \right\rceil$
2	2	1
3	2	1
3	3	1
4	2	2
4	3	1
4	4	1
5	2	2
5	3	2
5	4	1
5	5	1
6	2	2
6	3	2
6	4	2
6	5	1
6	6	1

(b) $7 \leq n \leq 10$

n	r	$\left\lceil \frac{\binom{n}{r}}{r(n-r)+1} \right\rceil$
7	2	2
7	3	3
7	4	3
7	5	2
7	6	1
7	7	1
8	2	3
8	3	4
8	4	5
8	5	4
8	6	3
8	7	1
8	8	1
9	2	3
9	3	5
9	4	6
9	5	6
9	6	5
9	7	3
9	8	1
9	9	1
10	2	3
10	3	6
10	4	9
10	5	10
10	6	9
10	7	6
10	8	3
10	9	1
10	10	1

Remark 1.1. The bound obtained above is an *expected lower bound* which we get from dimension count [Lan12]. It is not exact, since secant varieties of Grassmannians may have dimension smaller than their expected dimension (which is called *defective* cases).

Chapter 2

Theoretical Foundations and Computational Verification

2.1 Exterior Algebra and Alternating Tensors

2.1.1 Theory

[Win10, Gre67]

Let V be an n-dimensional vector space over a field $\mathbb{F}$ (In most case we work with $\mathbb{R}$).

Definition 2.1 (Tensor)**.** Let V and W be vector spaces over a field $\mathbb{F}$. The tensor product of V and W, denoted by

$$V \otimes W,$$

is the vector space generated by expression of form

$$v \otimes w, \qquad v \in V,\ w \in W,$$

such that they satisfies the biliear relations

$$(u+v) \otimes w = u \otimes w + v \otimes w,$$

$$u \otimes (v+w) = u \otimes v + u \otimes w,$$

and

$$(\lambda u) \otimes v = u \otimes (\lambda v) = \lambda(u \otimes v),$$

for every $u, v \in V$, $w \in W$, and $\lambda \in \mathbb{F}$.

An element of the tensor product space is called a *Tensor*.

Definition 2.2 (Higher-Order Tensor)**.** For an integer $k \geq 1$, the k-fold tensor product of V is

$$V^{\otimes k} = \underbrace{V \otimes V \otimes \cdots \otimes V}_{k \text{ factors}}.$$

The space $V^{\otimes k}$ is generated by expressions

$$v_1 \otimes v_2 \otimes \cdots \otimes v_k, \qquad v_i \in V,$$

satisfying the multilinearity relations

$$v_1 \otimes \cdots \otimes (u+v) \otimes \cdots \otimes v_k = v_1 \otimes \cdots \otimes u \otimes \cdots \otimes v_k + v_1 \otimes \cdots \otimes v \otimes \cdots \otimes v_k,$$

and

$$v_1 \otimes \cdots \otimes (\lambda u) \otimes \cdots \otimes v_k = \lambda \left(v_1 \otimes \cdots \otimes u \otimes \cdots \otimes v_k \right),$$

for every position, every $u, v \in V$, and every scalar $\lambda \in \mathbb{F}$.

An element of $V^{\otimes k}$ is called a *tensor of order k*. Every tensor can be written as a finite linear combination

$$T = \sum_{i=1}^{m} v_{i1} \otimes v_{i2} \otimes \cdots \otimes v_{ik},$$

where $v_{ij} \in V$.

Definition 2.3 (Alternating Tensor)**.** A tensor

$$T \in V^{\otimes k}$$

is called *alternating* if

$$T(v_1, \ldots, v_i, \ldots, v_j, \ldots, v_k) = -T(v_1, \ldots, v_j, \ldots, v_i, \ldots, v_k)$$

whenever we interchange the two arguments.

or we can also say that if two arguments are same then

$$T(v_1, \ldots, v, \ldots, v, \ldots, v_k) = 0.$$

The vector space consisting of all alternating tensors of order k is called the k-th exterior power of V.

Definition 2.4 (Exterior Power)**.** The exterior product of k copies of V, also called the k-th exterior power of V, is denoted $\Lambda^k(V)$.

It is defined as the subspace of totally antisymmetric tensors within

$$V^{\otimes k} = \underbrace{V \otimes \cdots \otimes V}_{k \text{ factors}}.$$

Its elements, called *alternating tensors* or *k-vectors*, are spanned by expressions of the form

$$v_1 \wedge v_2 \wedge \cdots \wedge v_k, \qquad v_i \in V,$$

where the wedge product satisfies multilinearity and antisymmetry.

With the help of alternating property we can say that

$$v_i \wedge v_j = -v_j \wedge v_i.$$

which means every wedge product containing repeated vectors vanishes.

Let

$$\mathcal{B} = \{e_1, e_2, \ldots, e_n\}$$

be an ordered basis of V. If wedge contain repeated basis vector it vansihed to zero ,we can reordered every wedge into increasing index order by using anticommutativity. Therefore every element of $\Lambda^k(V)$ can be expressed as a linear combination of wedges of the form

$$e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_k}, \qquad 1 \leq i_1 < i_2 < \cdots < i_k \leq n.$$

This leads to the standard basis of the exterior power.

Proposition 2.5. *If $\{e_1, \ldots, e_n\}$ is a basis of V, then*

$$\{e_{i_1} \wedge \cdots \wedge e_{i_k} : 1 \leq i_1 < \cdots < i_k \leq n\}$$

forms a basis of $\Lambda^k(V)$.

The above proposition immediately gives the dimension of the exterior power.

Theorem 2.6. *If $\dim(V) = n$, then*

$$\dim \Lambda^k(V) = \binom{n}{k}.$$

Proof. By the previous proposition, every basis element of $\Lambda^k(V)$ is uniquely determined by choosing an increasing k-tuple

$$(i_1, \ldots, i_k), \qquad 1 \leq i_1 < \cdots < i_k \leq n.$$

Conversely, every such choice produces a distinct basis element. □

Definition 2.7 (Decomposable Tensor)**.** A tensor

$$\omega \in \Lambda^k(V)$$

is called *decomposable* if there exist vectors $v_1, \ldots, v_k \in V$ such that

$$\omega = v_1 \wedge v_2 \wedge \cdots \wedge v_k.$$

If no such vectors exist, then ω is called *non-decomposable* (or *indecomposable*).

2.1.2 Computational Implementation

The basic operations of exterior algebra were implemented in Sagemath.

- `basis3`: Generates the ordered basis of $\Lambda^3(\mathbb{R}^6)$ consisting of all increasing triples (i, j, k), corresponding to the basis elements

$$e_i \wedge e_j \wedge e_k, \qquad i < j < k.$$

- `basis2`: Generates the ordered basis of $\Lambda^2(\mathbb{R}^6)$ consisting of all increasing pairs (i, j), corresponding to the basis elements

$$e_i \wedge e_j, \qquad i < j.$$

- `alpha_from_coeffs(coeffs)`: Constructs an alternating 3-vector from its 20 coordinate coefficients with respect to the basis of $\Lambda^3(\mathbb{R}^6)$.
- `alpha_from_wedge(*triples)`: Computes the wedge product of three vectors and returns the corresponding decomposable alternating 3-vector.
- `permutation_sign(seq)`: Computes the sign of a permutation required to preserve the antisymmetry of the wedge product.
- `wedge(A,B)`: Implements the wedge product of two alternating tensors, automatically enforcing antisymmetry and eliminating terms with repeated indices.

2.2 Decomposition in $\Lambda^2(\mathbb{R}^4)$

2.2.1 Theory

Necessary Condition for decomposability

Suppose that $\omega \in \Lambda^2(V)$ is decomposable. By definition, there exist vectors $x, y \in V$ such that:

$$\omega = x \wedge y$$

we compute the exterior product of ω with itself:

$$\omega \wedge \omega = (x \wedge y) \wedge (x \wedge y)$$

With properties of wedge product: $x \wedge x = 0$ and $y \wedge x = -x \wedge y$, it follows immediately that if ω is decomposable, it must satisfy:

$$\omega \wedge \omega = 0 \tag{2.1}$$

Thus, $\omega \wedge \omega = 0$ serves as a necessary condition for a 2-vector in a 4-dimensional space to be decomposable. **Any 2-vector for which $\omega \wedge \omega \neq 0$ is indecomposable.**

To find the explicit condition on the coefficients when the exterior product of a 2-vector with itself is zero ($\omega \wedge \omega = 0$).

Let the 2-vector ω be expressed using indexed coefficients:

$$\omega = a_{12}(e_1 \wedge e_2) + a_{13}(e_1 \wedge e_3) + a_{14}(e_1 \wedge e_4) + a_{23}(e_2 \wedge e_3) + a_{24}(e_2 \wedge e_4) + a_{34}(e_3 \wedge e_4)$$

we compute the expansion of $\omega \wedge \omega$. Any term that repeats a basis vector automatically becomes zero (for example, $e_1 \wedge e_2 \wedge e_1 \wedge e_3 = 0$ because e_1 is repeated).

Therefore, when we cross-multiply all the terms, the only combinations that remain non zero are pairs that share no indices at all. This leaves us with exactly 6 surviving terms:

$$\begin{aligned}
\omega \wedge \omega = {} & a_{12} \cdot a_{34}(e_1 \wedge e_2 \wedge e_3 \wedge e_4) \\
& + a_{13} \cdot a_{24}(e_1 \wedge e_3 \wedge e_2 \wedge e_4) \\
& + a_{14} \cdot a_{23}(e_1 \wedge e_4 \wedge e_2 \wedge e_3) \\
& + a_{23} \cdot a_{14}(e_2 \wedge e_3 \wedge e_1 \wedge e_4) \\
& + a_{24} \cdot a_{13}(e_2 \wedge e_4 \wedge e_1 \wedge e_3) \\
& + a_{34} \cdot a_{12}(e_3 \wedge e_4 \wedge e_1 \wedge e_2)
\end{aligned}$$

Next, we rearrange the basis vectors in every term back to the standard ordering ($e_1 \wedge e_2 \wedge e_3 \wedge e_4$). After balancing each rearrangement with change in sign we get

$$\omega \wedge \omega = 2 \cdot (a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23})(e_1 \wedge e_2 \wedge e_3 \wedge e_4)$$

Since the basis vectors are linearly independent their wedge product $(e_1 \wedge e_2 \wedge e_3 \wedge e_4)$ cannot be zero.

Thus, for $\omega \wedge \omega$ to equal the zero vector, the scalar part inside the parentheses must equal zero and if we assume that characterstic of field is not 2 then this gives us our necessary condition for a 2-vector to be decomposable:

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0 \tag{2.2}$$

Conversely, if this combination of coefficients does not equal zero, then $\omega \wedge \omega$ is not zero, which proves that the 2-vector is non-decomposable.

Sufficient Condition for decomposability in $\Lambda^2(V)$ with $\dim(V) = 4$

We now claim that $\omega \wedge \omega = 0$ is also a sufficient condition for the decomposability of alternating tensors

$$\omega = a_{12}(e_1 \wedge e_2) + a_{13}(e_1 \wedge e_3) + a_{14}(e_1 \wedge e_4) + a_{23}(e_2 \wedge e_3) + a_{24}(e_2 \wedge e_4) + a_{34}(e_3 \wedge e_4)$$

in $\Lambda^2(V)$ for a 4-dimensional vector space V. We assume $\omega \wedge \omega = 0$, which implies:

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0$$

To prove decomposability, we construct the factorization of ω under the following cases:

Case I

If $a_{12} = a_{13} = a_{23} = 0$, then the expression for ω reduces to:

$$\omega = a_{14}(e_1 \wedge e_4) + a_{24}(e_2 \wedge e_4) + a_{34}(e_3 \wedge e_4)$$

$$= (a_{14}e_1 + a_{24}e_2 + a_{34}e_3) \wedge e_4$$

Hence, ω is decomposable.

Case II

If at least one of a_{12}, a_{13}, a_{23} is non-zero, we analyze based on the value of a_{14}:

Sub-case 2.1: If $a_{14} \neq 0$

We assume that out of the leading coefficients $\{a_{12}, a_{13}, a_{23}\}$, the non-zero term is $a_{12} \neq 0$.

Given the relation:

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0$$

We begin with the full expression for the 2-vector:

$$\omega = a_{12}e_1 \wedge e_2 + a_{13}e_1 \wedge e_3 + a_{23}e_2 \wedge e_3 + a_{14}e_1 \wedge e_4 + a_{24}e_2 \wedge e_4 + a_{34}e_3 \wedge e_4$$

$$\Rightarrow e_1 \wedge (a_{12}e_2 + a_{13}e_3) + a_{23}e_2 \wedge e_3$$

$$= e_1 \wedge (a_{12}e_2 + a_{13}e_3) + \frac{a_{23}}{a_{12}}(a_{12}e_2 + a_{13}e_3) \wedge e_3$$

$$= \left(e_1 - \frac{a_{23}}{a_{12}}e_3\right) \wedge (a_{12}e_2 + a_{13}e_3)$$

$$\Rightarrow a_{14}e_1 \wedge e_4 + a_{24}e_2 \wedge e_4 + a_{34}e_3 \wedge e_4$$

$$= a_{14}e_1 \wedge e_4 + (a_{12}e_2 + a_{13}e_3) \wedge \frac{a_{24}}{a_{12}}e_4 + \left(a_{34} - \frac{a_{13}a_{24}}{a_{12}}\right) e_3 \wedge e_4$$

$$= \left(a_{14}e_1 + \frac{a_{34}a_{12} - a_{13}a_{24}}{a_{12}}e_3\right) \wedge e_4 + (a_{12}e_2 + a_{13}e_3) \wedge \frac{a_{24}}{a_{12}}e_4$$

From the initial condition $a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0$:

$$= \left(a_{14}e_1 - \frac{a_{14}a_{23}}{a_{12}}e_3\right) \wedge e_4 + (a_{12}e_2 + a_{13}e_3) \wedge \frac{a_{24}}{a_{12}}e_4$$

$$\Rightarrow \left(e_1 - \frac{a_{23}}{a_{12}}e_3\right) \wedge a_{14}e_4 + (a_{12}e_2 + a_{13}e_3) \wedge \frac{a_{24}}{a_{12}}e_4 + \left(e_1 - \frac{a_{23}}{a_{12}}e_3\right) \wedge (a_{12}e_2 + a_{13}e_3)$$

Let:

$$a = e_1 - \frac{a_{23}}{a_{12}}e_3$$

$$b = a_{12}e_2 + a_{13}e_3$$

$$x = e_4$$

$$c_1 = a_{14}$$

$$c_2 = \frac{a_{24}}{a_{12}}$$

$$= c_1 a \wedge x + c_2 b \wedge x + a \wedge b$$

$$= a \wedge (c_1 x + b) + (c_2 b \wedge x)$$

$$= (a + c_2 b) \wedge (c_1 x + b)$$

Substituting the expressions back:

$$\left(e_1 - \frac{a_{23}}{a_{12}}e_3 + \frac{a_{24}}{a_{12}a_{14}}(a_{12}e_2 + a_{13}e_3)\right) \wedge (a_{14}e_4 + a_{12}e_2 + a_{13}e_3)$$

$$\left(e_1 + \frac{a_{12}a_{24}}{a_{12}a_{14}}e_2 + \frac{a_{24}a_{13} - a_{23}a_{14}}{a_{12}a_{14}}e_3\right) \wedge (a_{14}e_4 + a_{12}e_2 + a_{13}e_3)$$

Using the identity $a_{24}a_{13} - a_{23}a_{14} = a_{12}a_{34}$:

$$\left(e_1 + \frac{a_{24}}{a_{14}}e_2 + \frac{a_{34}}{a_{14}}e_3\right) \wedge (a_{14}e_4 + a_{12}e_2 + a_{13}e_3)$$

Sub-case 2.2: If $a_{14} = 0$

When $a_{14} = 0$, the vanishing condition simplifies to $a_{12}a_{34} - a_{13}a_{24} = 0$. Depending on which of the leading coefficients is non-zero, we obtain the following structural factorizations:

1. If $a_{12} \neq 0$:

   We begin with the full expression for the 2-vector:

$$\omega = a_{12}e_1 \wedge e_2 + a_{13}e_1 \wedge e_3 + a_{23}e_2 \wedge e_3 + a_{24}e_2 \wedge e_4 + a_{34}e_3 \wedge e_4$$

$$\Rightarrow a_{12}e_1 \wedge e_2 + a_{13}e_1 \wedge e_3 + a_{23}e_2 \wedge e_3$$

$$\Rightarrow \left(e_1 - \frac{a_{23}}{a_{12}}e_3\right) \wedge (a_{12}e_2 + a_{13}e_3)$$

$$\Rightarrow a_{24}e_2 \wedge e_4 + a_{34}e_3 \wedge e_4$$

$$= (a_{12}e_2 + a_{13}e_3) \wedge \frac{a_{24}}{a_{12}}e_4$$

$$\Rightarrow \left(e_1 - \frac{a_{23}}{a_{12}}e_3\right) \wedge (a_{12}e_2 + a_{13}e_3) + (a_{12}e_2 + a_{13}e_3) \wedge \frac{a_{24}}{a_{12}}e_4$$

$$= (a_{12}e_2 + a_{13}e_3) \wedge \left(\frac{a_{24}}{a_{12}}e_4 + \frac{a_{23}}{a_{12}}e_3 - e_1\right)$$

   Similiarly we got :

2. If $a_{13} \neq 0$:

$$\omega = (a_{13}e_3 + a_{12}e_2) \wedge \left(\frac{a_{34}}{a_{13}}e_4 - \frac{a_{23}}{a_{13}}e_2 - e_1\right)$$

3. If $a_{23} \neq 0$:

   Since

$$a_{14} = a_{12} = a_{13} = 0,$$

   the bivector reduces to

$$\omega = a_{23}(e_2 \wedge e_3) + a_{24}(e_2 \wedge e_4) + a_{34}(e_3 \wedge e_4).$$

   We rewrite the first two terms as

$$a_{23}(e_2 \wedge e_3) + a_{24}(e_2 \wedge e_4) = e_2 \wedge (a_{23}e_3 + a_{24}e_4).$$

   Also,

$$a_{34}(e_3 \wedge e_4) = -\frac{a_{34}}{a_{23}}e_4 \wedge (a_{23}e_3 + a_{24}e_4),$$

since

$$e_4 \wedge e_4 = 0.$$

Therefore,

$$\begin{aligned}\omega &= e_2 \wedge (a_{23}e_3 + a_{24}e_4) - \frac{a_{34}}{a_{23}} e_4 \wedge (a_{23}e_3 + a_{24}e_4) \\ &= \left(e_2 - \frac{a_{34}}{a_{23}} e_4\right) \wedge (a_{23}e_3 + a_{24}e_4)\,.\end{aligned}$$

In every possible configuration of coefficients, ω can be written as a single-term exterior product of two vectors. Therefore, $\omega \wedge \omega = 0$ is a sufficient condition for decomposability.

Theorem 2.8. *A 2-vector $\omega \in \Lambda^2(V)$ in a 4-dimensional vector space V is decomposable if and only if:*

$$\omega \wedge \omega = 0$$

Consequently, ω is non-decomposable if and only if:

$$\omega \wedge \omega \neq 0$$

In terms of the component coefficients relative to a standard basis, the condition $\omega \wedge \omega = 0$ holds if and only if the coefficients satisfy the algebraic relation:

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0$$

where

$$\omega = a_{12}(e_1 \wedge e_2) + a_{13}(e_1 \wedge e_3) + a_{23}(e_2 \wedge e_3) + a_{14}(e_1 \wedge e_4) + a_{24}(e_2 \wedge e_4) + a_{34}(e_3 \wedge e_4).$$

2.2.2 Examples of Non-Decomposable 2-Vectors in $\Lambda^2(V)$ with $\dim(V) = 4$

Example 2.9. Consider $\omega = e_1 \wedge e_2 + e_3 \wedge e_4$. Here $a_{12} = 1, a_{34} = 1$, and all other coefficients are 0. Evaluating the condition:

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = (1)(1) - (0)(0) + (0)(0) = 1 \neq 0$$

Thus, $\omega \wedge \omega \neq 0$, making ω non-decomposable.

Example 2.10. Consider $\omega = 1(e_1 \wedge e_2) + 4(e_1 \wedge e_3) + 2(e_1 \wedge e_4) + 7(e_2 \wedge e_3) + 2(e_2 \wedge e_4) + 5(e_3 \wedge e_4)$. Evaluating the condition:

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = (1)(5) - (4)(2) + (2)(7) = 5 - 8 + 14 = 11 \neq 0$$

Thus, $\omega \wedge \omega \neq 0$, making ω non-decomposable.

We obtain the criteria to distinguishes decomposable and non-decomposable bivectors. However, our objective is stronger: to determine the smallest integer m such that every element of $\Lambda^2(\mathbb{R}^4)$ can be expressed as a sum of at most m decomposable bivectors.

To achieve our objective, we use the correspondence between bivectors and skew-symmetric bilinear forms. The following standard results from linear algebra provide the canonical forms required to classify $\Lambda^2(\mathbb{R}^4)$ under the action of $GL_4(\mathbb{R})$.

2.3 Linear Algebra Background

[HK71]

Theorem 2.11. *Let V be a finite-dimensional vector space over a field F. For each ordered basis $\mathcal{B}$ of V, the function which associates to each bilinear form on V its matrix in the ordered basis $\mathcal{B}$ is an isomorphism of the vector space $L(V, V; F)$ onto the space of $n \times n$ matrices over the field F.*

Theorem 2.12. *Let A and B be $n \times n$ matrices representing the same bilinear form on a vector space V in (possibly) different ordered bases. Then there exists an invertible $n \times n$ matrix P such that*

$$B = P^t A P.$$

Theorem 2.13. *Let V be an n-dimensional vector space over a subfield of the complex numbers, and let f be a skew-symmetric bilinear form on V. Then the rank r of f is even. If $r = 2k$, then there exists an ordered basis for V in which the matrix of f is the direct sum of the $(n-r) \times (n-r)$ zero matrix and k copies of the matrix*

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

That is,

$$[f]_{\mathcal{B}} = 0_{(n-r)\times(n-r)} \oplus \underbrace{J \oplus \cdots \oplus J}_{k \text{ copies}}.$$

2.4 Classification

If

$$\omega \in \Lambda^2(\mathbb{R}^4),$$

then with respect to the standard basis

$$e_1, e_2, e_3, e_4,$$

every 2-vector can be expanded as linear combination of basis as

$$\omega = a_{12} e_1 \wedge e_2 + a_{13} e_1 \wedge e_3 + a_{14} e_1 \wedge e_4 + a_{23} e_2 \wedge e_3 + a_{24} e_2 \wedge e_4 + a_{34} e_3 \wedge e_4.$$

We can arrange these coefficients in a 4×4 skew-symmetric matrix with

$$[A_\omega]_{ij} = a_{ij},$$

where

$$a_{ij} = -a_{ji}.$$

$$A_\omega = \begin{pmatrix} 0 & a_{12} & a_{13} & a_{14} \\ -a_{12} & 0 & a_{23} & a_{24} \\ -a_{13} & -a_{23} & 0 & a_{34} \\ -a_{14} & -a_{24} & -a_{34} & 0 \end{pmatrix}.$$

Definition 2.14. Let $\omega_1, \omega_2 \in \Lambda^2(\mathbb{R}^4)$ and let A_{ω_1} and A_{ω_2} be their corresponding skew-symmetric matrices.

We define relation $\sim$ on $\Lambda^2(\mathbb{R}^4)$ by

$$\omega_1 \sim \omega_2$$

if there exists an invertible matrix 4×4 matrix P such that

$$A_{\omega_1} = P^T A_{\omega_2} P.$$

We now going to prove that it is equivalnce relation.

Reflexivity. Let $\omega \in \Lambda^2(\mathbb{R}^4)$.

Take $P = I_4$, we have

$$A_\omega = I_4^T A_\omega I_4 = A_\omega.$$

Hence $\omega \sim \omega$. □

Symmetry. Suppose $\omega_1 \sim \omega_2$. then there exists a invertible matrix P such that

$$A_{\omega_1} = P^T A_{\omega_2} P.$$

Since P is invertible,

$$A_{\omega_2} = (P^{-1})^T A_{\omega_1} P^{-1}.$$

Therefore $\omega_2 \sim \omega_1$. □

Transitivity. Suppose

$$\omega_1 \sim \omega_2 \quad \text{and} \quad \omega_2 \sim \omega_3.$$

then there exist invertible matrices P, Q such that

$$A_{\omega_1} = P^T A_{\omega_2} P$$

and

$$A_{\omega_2} = Q^T A_{\omega_3} Q.$$

Substituting A_{ω_3} into the first equation, we get

$$A_{\omega_1} = P^T (Q^T A_{\omega_3} Q) P = (QP)^T A_{\omega_3} (QP).$$

Since QP is invertible, it follows that

$$\omega_1 \sim \omega_3.$$

Hence $\sim$ is transitive. □

Let $\omega \in \Lambda^2(\mathbb{R}^4)$ be a non-decomposable bivector. Then

$$\omega \wedge \omega \neq 0.$$

Since

$$\omega \wedge \omega = 2(a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23})\, e_1 \wedge e_2 \wedge e_3 \wedge e_4,$$

it follows that

$$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} \neq 0.$$

Hence

$$\det(A_\omega) = (a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23})^2 \neq 0.$$

Therefore A_ω is invertible and

$$\operatorname{rank}(A_\omega) = 4.$$

By Theorem 2.13, there exists a basis of $\mathbb{R}^4$ in which the matrix of ω is

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Equivalently, there exists an invertible matrix $P \in GL_4(\mathbb{R})$ such that

$$A_\omega = P^T \left(\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \oplus \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \right) P.$$

Consequently, every non-decomposable bivector is equivalent to

$$e_1 \wedge e_2 + e_3 \wedge e_4.$$

Hence all non-decomposable bivectors belong to a single equivalence class under the relation $\sim$.

On the other hand, every decomposable bivector can be written in the form

$$u \wedge v,$$

for some linearly independent vectors $u, v \in \mathbb{R}^4$.

Therefore, every bivector in $\Lambda^2(\mathbb{R}^4)$ belongs to exactly one of the following two classes:

$$\omega \sim u \wedge v, \qquad \text{or} \qquad \omega \sim e_1 \wedge e_2 + e_3 \wedge e_4.$$

Thus every element of $\Lambda^2(\mathbb{R}^4)$ is either decomposable or equivalent to the unique non-decomposable normal form. Consequently, every bivector can be expressed as the sum of at most two decomposable bivectors.

2.4.1 Computational Verification

We develop decomposability criterion in the previous section and verify it computationally.

1. `decompose(w)`: Given a bivector

   $$\omega = a_{12}e_{12} + a_{13}e_{13} + a_{14}e_{14} + a_{23}e_{23} + a_{24}e_{24} + a_{34}e_{34},$$

   this checks the whether bivector is decomposabile

   $$a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0.$$

   If this condition fails, the bivector is non-decomposable.

2. `verify_decompose(w)`: This function is create to validates the decomposition returned by `decompose`. It reconstructs the bivector

   $$\omega' = u \wedge v,$$

   using the implemented wedge product and compares the coefficients of ω' with those of the original input bivector. The decomposition is accepted only if all corresponding coefficients agree exactly, thereby providing an verification of proof.

2.5 Plücker Relations and Decomposability

2.5.1 Theory

In Section 2.2 we established that for 2-vectors in $\Lambda^2(\mathbb{R}^4)$ the quadratic condition

$$\omega \wedge \omega = 0$$

is both necessary and sufficient for decomposability. This criterion is a special case of a much more general characterization valid for arbitrary exterior powers. For $\Lambda^r(\mathbb{R}^n)$, decomposability is completely determined by a family of homogeneous quadratic equations known as the *Plücker relations*.

Let

$$\omega \in \Lambda^3(\mathbb{R}^6)$$

be written with respect to the standard basis as

$$\omega = \sum_{1 \le i < j < k \le 6} p_{ijk}\, e_i \wedge e_j \wedge e_k,$$

where the coefficients p_{ijk} are called the *Plücker coordinates* of ω.

For every choice of index sets

$$\{i_1, i_2\}, \qquad \{j_1, j_2, j_3, j_4\} \subseteq \{1, \dots, 6\},$$

the Plücker coordinates satisfy the quadratic relation

$$\sum_{t=1}^{4} (-1)^t\, p_{i_1 i_2 j_t}\, p_{j_1 \cdots \widehat{j_t} \cdots j_4} = 0,$$

where the symbol $\widehat{j_t}$ indicates that the index j_t is omitted.

Theorem 2.15 (Plücker Relations)**.** *A nonzero element*

$$\omega \in \Lambda^3(\mathbb{R}^6)$$

is decomposable if and only if its Plücker coordinates satisfy all of the Plücker relations. [Har92] Equivalently, the projective point

$$[\omega] \in \mathbb{P}\big(\Lambda^3(\mathbb{R}^6)\big)$$

lies on the Grassmannian

$$G(3, \mathbb{R}^6) \subset \mathbb{P}\big(\Lambda^3(\mathbb{R}^6)\big)$$

under the Plücker embedding if and only if ω is decomposable.

This theorem provide us a complete algebraic characterization of decomposable alternating tensors. It is already well-established proven theorem so we avoid to replicate the proof here, we use it as the theoretical basis for our computational algorithms. Our implementation checks whether all quadratic Plücker relations vanish and thereby determines whether a given alternating 3-vector is decomposable.

2.5.2 Computational Verification

To verify decomposability computationally, we implemented the function

$$\texttt{plucker_relations_3_6}(\omega),$$

which evaluates every quadratic Plücker relation for a given element

$$\omega \in \Lambda^3(\mathbb{R}^6).$$

The function returns the collection of relations that fail to vanish. Using this routine, we defined

$$\texttt{is_decomposable}(\omega),$$

which returns `True` precisely when no Plücker relation is violated.

We want to see how frequently decomposable tensors occur, we generated 1000 random alternating 3-vectors by choosing each of the twenty Plücker coordinates independently and uniformly from the set

$$\{-5, -4, \ldots, 5\}.$$

we find that none of the sampled tensors satisfied all Plücker relations; consequently, none were decomposable.

This behaviour agrees with the geometric dimension count obtained in Section 1.5. Indeed,

$$\dim G(3,6) = 3(6-3) = 9,$$

whereas

$$\dim \mathbb{P}\big(\Lambda^3(\mathbb{R}^6)\big) = 20 - 1 = 19.$$

Thus the Grassmannian of decomposable 3-vectors is a proper 9-dimensional algebraic subvariety inside a 19-dimensional projective space, so a randomly chosen element is expected to be non-decomposable with overwhelming probability.

We go further to see the behaviour of the Plücker relations under changes of basis. We apply random invertible matrices to change the basis from $\mathrm{GL}(6,\mathbb{R})$ theory showed that decomposability is preserved under change of basis but we find that the *number* of violated Plücker relations is not preserved. For example, one randomly generated tensor violated twelve Plücker relations, whereas after we change of basis through invertible matrix then the transformed tensor violated fifteen relations.

This observation show us that individual Plücker equations depend on the chosen coordinates. Only their common zero locus, corresponding to decomposable tensors and it is intrinsic under the action of $\mathrm{GL}(6,\mathbb{R})$. Hence the number of violated relations cannot be used as an invariant for orbit classification.

This motivated the search for genuine $\mathrm{GL}(6,\mathbb{R})$-invariants. In the next section we introduce the contraction map, whose rank remains unchanged under every invertible change of basis and provides a natural invariant for the classification of alternating 3-vectors.

2.5.3 Alternating 3-Vectors in $\Lambda^3(\mathbb{R}^6)$

Theory

Let

$$\omega \in \Lambda^3(V), \qquad \dim V = 6.$$

For every vector $v \in V$, the interior product (contraction)

$$v \lrcorner \omega \in \Lambda^2(V)$$

defines the linear map

$$\phi_\omega : V \longrightarrow \Lambda^2(V), \qquad \phi_\omega(v) = v \lrcorner \omega.$$

Since

$$\dim \Lambda^2(V) = \binom{6}{2} = 15,$$

the map is represented by a 15×6 matrix after choosing a basis.

Under the natural action of $GL(V)$,

$$\omega \mapsto g \cdot \omega,$$

the contraction map transforms by conjugation, and hence

$$\operatorname{rank}(\phi_\omega)$$

is invariant under change of basis. Consequently, the contraction rank provides a first invariant for classifying alternating 3-vectors.

Theorem 2.16. *[Bry06] Let V be a real vector space of dimension 6 and let $\phi \in \Lambda^3(V^*)$.*

Then every 3-form is $GL(V)$-equivalent to exactly one of the following six normal forms:

(1) $e^{123} + e^{456}$,
(2) $e^{135} - e^{146} - e^{236} - e^{245}$,
(3) $e^{156} + e^{264} + e^{345}$,
(4) $e^{125} + e^{345}$,
(5) e^{123},
(6) 0.

Moreover, these six forms are mutually inequivalent under the action of $GL(V)$. The first two orbits are open in $\Lambda^3(V^*)$, the third orbit is a hypersurface, and the remaining three have higher codimension.

Remark 2.17. Vectors versus Forms, and Notation]

Bryant's theorem is originally stated for alternating 3-forms $\varphi \in \Lambda^3(V^*)$ using the dual basis $e^1, \ldots, e^6$.

Throughout this report we work instead with alternating 3-vectors $\omega \in \Lambda^3(V)$ and write $e_i \wedge e_j \wedge e_k$. After fixing a basis of V, these two descriptions are naturally identified, so Bryant's orbit classification applies equally in our notation.

Accordingly, we write the six normal forms as

$$e_1\wedge e_2\wedge e_3+e_4\wedge e_5\wedge e_6, \quad e_1\wedge e_3\wedge e_5-e_1\wedge e_4\wedge e_6-e_2\wedge e_3\wedge e_6-e_2\wedge e_4\wedge e_5, \quad e_1\wedge e_5\wedge e_6+e_2\wedge e_6\wedge e_4+e_3\wedge e_4\wedge e_5,$$

$$e_1\wedge e_2\wedge e_5 + e_3\wedge e_4\wedge e_5, \qquad e_1\wedge e_2\wedge e_3, \qquad 0,$$

throughout the remainder of the report.

Bryant's theorem reduces the classification problem to identifying the $GL(6,\mathbb{R})$-orbit of a given alternating 3-vector.

Instead of attempting to compute an explicit change of basis sending a form to one of Bryant's representatives, we will compute invariants that distinguish the six orbit types.

The first invariant is the contraction rank $\operatorname{rank}(\phi_\omega)$.

If the contraction rank is strictly smaller than 6, the form is degenerate. Following Bryant's proof, degenerate forms admit a linear factor

$$\omega = \alpha \wedge \beta,$$

and are further distinguished by the rank of the associated skew-symmetric matrix of β.

If $\operatorname{rank}(\phi_\omega) = 6$, the form is non-degenerate. To distinguish the remaining three non-degenerate orbits we compute Hitchin's quartic invariant.[Hit00]

For non-degenerate forms we use Hitchin's operator

$$K_\omega : V \to V,$$

defined by

$$K_\omega(v) = A((v \lrcorner \omega) \wedge \omega),$$

where the volume form identifies $\Lambda^5(V^*) \cong V^*$.

The associated quartic invariant is

$$\lambda(\omega) = \frac{1}{6}\operatorname{tr}(K_\omega^2).$$

The sign of $\lambda(\omega)$ is invariant under the action of $GL(6,\mathbb{R})$ and distinguishes the two open Bryant orbits, while $\lambda(\omega) = 0$ characterizes the hypersurface orbit.

When we combine Bryant's normal forms with the contraction rank and Hitchin's invariantwe get the complete decision procedure to classify 3 vector implemented in this work.

$$\boxed{\omega \longrightarrow \operatorname{rank}(\phi_\omega) \longrightarrow \begin{cases} \text{Degenerate classifier}, & \operatorname{rank} < 6, \\ \lambda(\omega), & \operatorname{rank} = 6, \end{cases}}$$

which identifies the unique $GL(6,\mathbb{R})$-orbit of every alternating 3-vector.

Classification Strategy

The classification procedure implemented in this work is summarized in Figure 2.1.

$$\omega \in \Lambda^3(\mathbb{R}^6)$$

$$\downarrow$$

$$\text{Compute } \operatorname{rank}(\phi_\omega)$$

$$\downarrow$$

Degenerate	**Non-degenerate**
$\operatorname{rank}(\phi_\omega) < 6$	$\operatorname{rank}(\phi_\omega) = 6$
$\downarrow$	$\downarrow$
Find $\omega = \alpha \wedge \beta$	Compute $\lambda(\omega)$
$\downarrow$	$\downarrow$
Compute $\operatorname{rank}(\beta)$	Determine $\operatorname{sign}(\lambda)$
$\downarrow$	$\downarrow$
$\operatorname{rank}(\beta) = 0$	$\lambda(\omega) > 0$
$\implies 0$	$\implies e^{123} + e^{456}$
$\operatorname{rank}(\beta) = 2$	$\lambda(\omega) < 0$
$\implies e^{123}$	$\implies e^{135} - e^{146} - e^{236} - e^{245}$
$\operatorname{rank}(\beta) = 4$	$\lambda(\omega) = 0$
$\implies e^{125} + e^{345}$	$\implies e^{156} + e^{264} + e^{345}$

Figure 2.1: Classification procedure for alternating 3-vectors in $\Lambda^3(\mathbb{R}^6)$ based on Bryant's normal forms.

The first step is to determine whether the alternating 3-vector is degenerate. If

$$\operatorname{rank}(\phi_\omega) < 6,$$

then Bryant proves that ω admits a linear factorization

$$\omega = \alpha \wedge \beta,$$

where $\alpha \in \Lambda^1(\mathbb{R}^6)$ and $\beta \in \Lambda^2(\mathbb{R}^5)$. Consequently, the classification problem reduces to the classification of the alternating 2-form β. Since a skew-symmetric bilinear form on a 5-dimensional vector space has rank 0, 2, or 4, the rank of β completely determines the corresponding Bryant orbit.

If

$$\operatorname{rank}(\phi_\omega) = 6,$$

then ω is non-degenerate. In this case the contraction rank no longer distinguishes the remaining orbits. Bryant shows that the non-degenerate forms are classified by Hitchin's quartic invariant

$$\lambda(\omega) = \frac{1}{6}\operatorname{tr}(K_\omega^2).$$

The sign of $\lambda(\omega)$ determines the orbit completely: $\lambda(\omega) > 0$ and $\lambda(\omega) < 0$ correspond to the two open $GL(6,\mathbb{R})$-orbits, while $\lambda(\omega) = 0$ corresponds to the remaining hypersurface orbit.

Computational Verification

The classification algorithm was implemented in SageMath as a collection of functions corresponding to the mathematical operations appearing in Bryant's classification theorem.

- `contraction(v,ω)`: Computes the interior product

  $$i_v\omega \in \Lambda^2(\mathbb{R}^6),$$

  which is the fundamental operation used in constructing the contraction map.

- `phi_matrix(ω)`: Constructs the matrix representation of

  $$\phi_\omega : V \to \Lambda^2(V), \qquad \phi_\omega(v) = i_v\omega,$$

  with respect to the standard bases of V and $\Lambda^2(V)$.

- `phi_rank(ω)`: Computes the invariant

  $$\operatorname{rank}(\phi_\omega),$$

  used to distinguish degenerate and non-degenerate 3-vectors.

- `is_degenerate(ω)`: Tests the condition

  $$\operatorname{rank}(\phi_\omega) < 6,$$

  which determines whether ω is degenerate.

- `linear_factor_candidates(ω)`: Computes vectors

  $$\alpha \in V$$

  satisfying

  $$\omega = \alpha \wedge \beta,$$

  whenever ω is degenerate.

- `solve_beta(ω, α)`: Constructs the associated alternating 2-form

  $$\beta \in \Lambda^2(\mathbb{R}^5)$$

  from the decomposition

  $$\omega = \alpha \wedge \beta.$$

- `beta_rank(`β`)`: Computes the rank of the skew-symmetric matrix associated with β, namely
$$\operatorname{rank}(B_\beta),$$
which determines the degenerate Bryant normal form.

- `classify_degenerate(`ω`)`: Identifies the Bryant orbit of a degenerate form using
$$\operatorname{rank}(\beta) \in \{0, 2, 4\}.$$

- `compute_hitchin_operator(`ω`)`: Constructs Hitchin's operator
$$K_\omega(v) = A\big((v \lrcorner \omega) \wedge \omega\big),$$
used to compute the quartic invariant.

- `hitchin_lambda(`ω`)`: Computes Hitchin's invariant
$$\lambda(\omega) = \frac{1}{6} \operatorname{tr}(K_\omega^2),$$
whose sign distinguishes the three non-degenerate Bryant orbits.

- `classify(`ω`)`: Combines
$$\operatorname{rank}(\phi_\omega), \qquad \operatorname{rank}(\beta), \qquad \lambda(\omega)$$
to determine the $\mathrm{GL}(6, \mathbb{R})$-orbit of the input alternating 3-vector.

2.6 Exploratory Study on $\Lambda^3(\mathbb{R}^7)$

2.6.1 Theory

- For $\omega \in \Lambda^3(\mathbb{R}^7)$, the contraction map
$$\phi_\omega : \mathbb{R}^7 \longrightarrow \Lambda^2(\mathbb{R}^7), \qquad \phi_\omega(v) = \iota_v \omega,$$
is represented, after choosing a basis, by a 21×7 matrix, since $\dim \Lambda^2(\mathbb{R}^7) = \binom{7}{2} = 21$. As in the $\Lambda^3(\mathbb{R}^6)$ case, $\operatorname{rank}(\phi_\omega)$ is $GL(7, \mathbb{R})$-invariant, so it is a natural first invariant to compute; by the shape of the matrix it is bounded above by 7.

  Unlike the $\Lambda^3(\mathbb{R}^6)$ case, we did not use a complete orbit classification here. From Table 1.1, the expected lower bound on the number of decomposable summands for $(n, r) = (7, 3)$ is $m = 3$.

  Each contraction $\iota_{e_i}\omega \in \Lambda^2(\mathbb{R}^7)$, for $i = 1, \ldots, 7$, is itself a skew-symmetric 7×7 matrix under the standard identification of $\Lambda^2(\mathbb{R}^7)$ with $\mathrm{Skew}_7(\mathbb{R})$. The 21×7 matrix of ϕ_ω is exactly these seven skew-symmetric matrices, each flattened into its 21 independent entries and placed as one column; this is what the `contraction` and `contraction_rank` functions build and take the rank of.

- During the computational study of alternating 3-vectors, the repeated use of the contraction map naturally led to the correspondence between bivectors and skew-symmetric matrices, also every element

$$\omega \in \Lambda^2(\mathbb{R}^n)$$

can be written uniquely as

$$\omega = \sum_{i<j} a_{ij}\, e_i \wedge e_j,$$

and is naturally identified with the skew-symmetric matrix

$$A_\omega = (a_{ij}), \qquad a_{ji} = -a_{ij}.$$

Under this identification, decomposable bivectors correspond to skew-symmetric matrices of rank 2 (or rank 0 in the trivial case), while the matrix rank provides a natural invariant for their classification.

This viewpoint motivated an exploratory reading of[KK25], which studies alternating bilinear maps in a more general setting. Although the results of that paper are not used directly in the classification algorithms developed in this report, the correspondence between $\Lambda^2(\mathbb{R}^n)$, skew-symmetric matrices, and alternating bilinear maps provides a natural conceptual connection with the computational framework used throughout this work. This connection suggests a possible direction for extending the present work, and forms part of the ongoing research direction.

2.6.2 Computational Verification

We tested five explicit alternating 3-vectors of increasing complexity: sums of 1, 2, or 3 decomposable terms with varying degrees of shared indices. The single decomposable term $e_0 \wedge e_1 \wedge e_2$ gave contraction rank 3 and was confirmed decomposable, as expected. The sum of two disjoint decomposable terms ($e_0 \wedge e_1 \wedge e_2 + e_3 \wedge e_4 \wedge e_5$) gave contraction rank 6 and was non-decomposable. Two further examples, each a sum of two decomposable terms sharing a single common basis index, both gave contraction rank 5 and were also non-decomposable. A sum of three terms gave the maximum possible contraction rank 7, again non-decomposable. In every non-decomposable case, the decomposability check used throughout this work (built on the Plücker-relation test) returned false; we have not separately re-verified here which individual Plücker relations were violated.

Sampling 200 random alternating 3-vectors with integer coefficients in $\{-5, \ldots, 5\}$, every sample returned contraction rank 7 and was non-decomposable — consistent with decomposable and low-rank tensors forming a lower-dimensional subvariety, as in the $\Lambda^3(\mathbb{R}^6)$ case.

To look more closely at the rank spectrum, we ran a search over 50,000 random tensors with coefficients in $\{-2, \ldots, 2\}$, checking specifically for contraction rank 4. None was found. This is an empirical observation, not a proof: the search was limited in range and did not separately probe ranks 0, 1, or 2. Settling which ranks actually occur would need either a theoretical argument or a larger, more structured search.

```
# Can rank 4 occur? Try systematically
for i in range(50000):
    w = {b: randint(-2,2) for b in basis}
    w = {k:v for k,v in w.items() if v != 0}
    if w:
        r = contraction_rank(w)
        if r == 4:
            print("Found rank 4!", w)
            break
else:
    print("Rank 4 never occurs")
```

```
Out[11]: Rank 4 never occurs
```

Figure 2.2: SageMath experiment searching for an alternating 3-vector in $\Lambda^3(\mathbb{R}^7)$ with contraction rank 4. Among 50,000 randomly generated tensors with coefficients in $\{-2, \ldots, 2\}$, no example of contraction rank 4 was found.

2.7 Discussion

2.7.1 Observations

The dimension-counting argument of Section 1.5 correctly predicts the near-total scarcity of decomposable tensors for $\Lambda^3(\mathbb{R}^6)$: no randomly sampled tensor out of 1000 satisfied all Plücker relations, matching the fact that the 9-dimensional Grassmannian $G(3,6)$ sits properly inside the 19-dimensional projective space $\mathbb{P}(\Lambda^3(\mathbb{R}^6))$. A more subtle finding is that while the individual Plücker relations are not $GL(6,\mathbb{R})$-invariant (the number of violated relations changed from 12 to 15 under a single change of basis), the contraction rank $\text{rank}(\phi_\omega)$ remained fixed under the same transformation, confirming it as the correct invariant to build the classifier around. The Hitchin decomposition experiment further suggests that forms in the hyperbolic orbit ($\lambda(\omega) < 0$) carry an intrinsic complex structure, splitting canonically into a conjugate pair of decomposable complex 3-forms.

```
# Take a vector with 12 violations and apply a random GL6 transformation
# Does the violation count change?

import numpy as np

def apply_gl6(omega_dict, M):
    """Apply GL6 matrix M to a 3-vector omega."""
    result = {}
    for i,j,k in combinations(range(6), 3):
        val = 0
        for a,b,c in combinations(range(6), 3):
            # M maps e_a -> sum_i M[i,a] e_i
            val += omega_dict.get((a,b,c), 0) * np.linalg.det(M[np.ix_([i,j,k],
[a,b,c])])
        if abs(val) > 1e-10:
            result[(i,j,k)] = val
    return result

# Find a 12-violation vector
for _ in range(10000):
    omega = {b: random.randint(-5,5) for b in basis}
    v = plucker_relations_3_6(omega)
    if len(v) == 12:
        omega_12 = omega
        break

# Apply random invertible matrix
M = np.random.randn(6,6)
while abs(np.linalg.det(M)) < 0.1:
    M = np.random.randn(6,6)

omega_transformed = apply_gl6(omega_12, M)
print("Before:", len(plucker_relations_3_6(omega_12)))
print("After GL6:", len(plucker_relations_3_6(omega_transformed)))
```

```
Out[5]: Before: 12
        After GL6: 15
```

(a) The number of plucker violation goes from 12 to 15 under change of basis.

```
basis_2 = list(combinations(range(6), 2))  # 15 basis elements of Lambda^2
basis_3 = list(combinations(range(6), 3))  # 20 basis elements of Lambda^3

def contraction_matrix(omega):
    """
    Build the 15x6 matrix of the contraction map v -> v ⌟ omega.
    Rows: basis elements of Lambda^2 (15 of them)
    Cols: basis vectors e_0,...,e_5 (6 of them)
    Entry (jk, i) = omega_{ijk} with appropriate sign
    """
    M = np.zeros((15, 6))
    for col, i in enumerate(range(6)):
        for row, (j,k) in enumerate(basis_2):
            if i == j or i == k:
                continue
            triple = tuple(sorted([i,j,k]))
            # sign of permutation (i,j,k) -> sorted
            perm = [i,j,k]
            sign = wedge_sign(perm)  # your existing function
            M[row, col] = sign * omega.get(triple, 0)
    return M

def contraction_rank(omega):
    M = contraction_matrix(omega)
    return np.linalg.matrix_rank(M, tol=1e-9)


from collections import Counter
ranks = []
for _ in range(1000):
    omega = {b: random.randint(-5,5) for b in basis}
    ranks.append(contraction_rank(omega))
print(Counter(ranks))


print("Before:", contraction_rank(omega_12))
print("After GL6:", contraction_rank(apply_gl6(omega_12, M)))
```

```
Out[7]: Counter({np.int64(6): 1000})
        Before: 6
        After GL6: 6
```

(b) Contraction rank before and after a random $GL(6,\mathbb{R})$ transformation remains unchanged.

Figure 2.3: Computational verification of the contraction rank invariant.

2.7.2 Computational Challenges

The main computational challenge was distinguishing genuine $GL(6, \mathbb{R})$-invariants from coordinate-dependent quantities; the Plücker violation count initially appeared to be a natural classifying statistic but was shown by direct experiment to fail this requirement. A second challenge was numerical: the Hitchin decomposition involves square roots of $-\lambda(\omega)$ and complex arithmetic, requiring care to keep reconstruction error within machine precision across widely varying magnitudes of $\lambda(\omega)$ (from -4 to -81 in the tested cases).Third challenge i face with is during arithmetics because the implementation uses floating-point arithmetic and exact zeros may be represented by very small numerical values. Throughout the computations, a tolerance $\varepsilon = 10^{-12}$ was used, and values satisfying $|x| < \varepsilon$ were regarded as zero.

2.7.3 Limitations

The classification of $\Lambda^3(\mathbb{R}^6)$ implemented in this report is based on Bryant's classification theorem, which is used as a theoretical input rather than derived from first principles. The computational experiments serve to verify the implementation and illustrate the theory; they do not constitute mathematical proofs.

The exploratory study of $\Lambda^3(\mathbb{R}^7)$ is based on randomly generated alternating 3-vectors with coefficients chosen from a finite range. Such random sampling is useful for identifying patterns, but it does not sample all orbit types uniformly. Consequently, observations such as the apparent absence of contraction rank 4 should be regarded as experimental evidence not as mathematical results.

Chapter 3

Conclusion and Future Work

3.1 Summary

I work on a theoretical and computational implementation of decomposability and orbit classification of alternating tensors within exterior algebra, focusing on low-dimensional vector spaces. By evaluating the secant varieties of Grassmannians $G(r, \mathbb{R}^n)$ embedded inside the projective space $\mathbb{P}(\Lambda^r\mathbb{R}^n)$, the work establishes an expected lower bound for the minimal number of decomposable summands required to express any r-vector as $m = \lceil \frac{\binom{n}{r}}{r(n-r)+1} \rceil$. For $\Lambda^2(\mathbb{R}^4)$, the necessary and sufficient algebraic condition for a bivector to be decomposable is the quadratic relation $\omega \wedge \omega = 0$, which evaluates to the standard Plücker relation $a_{12}a_{34} - a_{13}a_{24} + a_{14}a_{23} = 0$. Representing these bivectors as 4×4 skew-symmetric matrices reveals an equivalence relation under the action of the general linear group $GL_4(\mathbb{R})$, confirming that all non-decomposable elements fall into a single orbit with the normal form $e_1 \wedge e_2 + e_3 \wedge e_4$, bound by a maximum tensor rank of 2 .

For higher-order tensors in $\Lambda^3(\mathbb{R}^6)$, where the Grassmannian forms a 9-dimensional subvariety within a 19-dimensional projective space, random sampling confirms that generic 3-vectors are non-decomposable with overwhelming probability . The report applies Bryant's classification theorem to systemize these 3-vectors into six mutually inequivalent $GL_6(\mathbb{R})$-orbits . To algorithmically identify these orbits without explicitly computing a change of basis, a progressive decision tree is implemented using coordinate-independent invariants . Degenerate forms are isolated when the linear contraction map $\phi_\omega(v) = v \lrcorner \omega$ has a rank strictly less than 6, reducing the problem to a secondary factorization $\omega = \alpha \wedge \beta$ classified by $\text{rank}(\beta) \in \{0, 2, 4\}$. For non-degenerate forms where $\text{rank}(\phi_\omega) = 6$, Hitchin's quartic invariant $\lambda(\omega) = \frac{1}{6}\text{tr}(K_\omega^2)$ is evaluated; its sign distinguishes the two open orbits ($\lambda(\omega) > 0$ and $\lambda(\omega) < 0$) from the remaining singular hypersurface orbit ($\lambda(\omega) = 0$) .

With this , the decomposition problem posed in Section 1.2 is solved for $\Lambda^2(\mathbb{R}^4)$ and $\Lambda^3(\mathbb{R}^6)$ where the minimal values of m are 2 and 3, respectively. The corresponding question for higher-dimensional exterior powers is what i'll investigate in future.

3.2 Future Directions

- I want to extend the computational study of alternating 3-forms to higher-dimensional exterior spaces such as $\Lambda^3(\mathbb{R}^7)$ and investigate whether similar classification techniques can be developed.

- I will try to generalize the constructive decomposition methods developed for $\Lambda^2(\mathbb{R}^4)$ and $\Lambda^3(\mathbb{R}^6)$ to higher-dimensional exterior powers.
- I have decided to explore the recently proposed graph-theoretic framework that associates alternating bilinear equations with weighted graphs and investigate how graph properties can help in of alternating bilinear equations.[KK25]

Appendix A

Algorithms

Computational work of this project was implemented using **SageMath**.

I list down principal functions in this chapter. Many auxiliary functions used for input handling, basis generation, testing, verification, visualization, and experimental computations have been omitted to keep this section simple but it is available with my complete work during this project in GitHub repository:

https://github.com/akarsh-suryavanshi/mathematical_computing/tree/main/Summer_Internship_IISER_Mohali_2026

Main function together with their mathematical purpose are as follow :

General Exterior Algebra functions

- `permutation_sign()` – Computes the sign of a permutation.
- `wedge_product()` – Computes the wedge product of vectors.
- `wedge()` – Computes the exterior product of alternating forms.
- `plucker()` – Evaluates the Plücker relations.
- `is_decomposable()` – Tests decomposability using the Plücker relations.
- `contraction()` – Computes the interior product.
- `contraction_rank()` – Computes the contraction rank.

Algorithms for $\Lambda^2(\mathbb{R}^4)$

- `decompose()` – Computes an explicit decomposition $\omega = u \wedge v$.
- `verify_decompose()` – Verifies the computed decomposition.

Hitchin Theory

- `hitchin_operator()` – Constructs Hitchin's operator K_ω.
- `hitchin_lambda()` – Computes Hitchin's quartic invariant.
- `stable_type()` – Determines whether a form is positive, negative, or degenerate.

- `hitchin_star_alpha()` – Computes the image of a 3-form under Hitchin's operator.
- `hitchin_decomposition()` – Computes the decomposition of positive stable forms.
- `hitchin_decomposition_negative()` – Computes the complex decomposition of negative stable forms.
- `contraction_form()` – Computes the contraction of a 3-form as a 2-form.
- `beta_s()` – Computes $(i_{e_s}\omega) \wedge \omega$ used in constructing Hitchin's operator.
- `fiveform_to_vector()` – Converts a 5-form into a vector via the volume form.

Bryant Classification of $\Lambda^3(\mathbb{R}^6)$

- `alpha_from_coeffs()` – Constructs a 3-form from its coordinate vector.
- `alpha_from_wedge()` – Constructs a 3-form as a sum of decomposable forms.
- `phi_matrix()` – Constructs the matrix of the contraction map.
- `phi_rank()` – Computes the contraction rank.
- `is_degenerate()` – Detects degenerate alternating 3-forms.
- `linear_factor_candidates()` – Computes candidate linear factors.
- `solve_beta()` – Solves $\omega = \alpha \wedge \beta$.
- `beta_rank()` – Computes the rank of the associated skew-symmetric matrix.
- `classify_degenerate()` – Classifies degenerate Bryant orbits.
- `classify_nondegenerate()` – Classifies non-degenerate forms using Hitchin's invariant.
- `classify_full()` – Returns the complete Bryant orbit of the input 3-form.
- `analyze()` – Displays the contraction matrix and its rank.
- `classify()` – Determines whether the input form is degenerate or non-degenerate.

Testing and Verification

- `random_3form()` – Generates random alternating 3-forms for experimental testing.
- `verify_decompose()` – Verifies the correctness of the computed decomposition.
- `analyze()` – Displays the contraction matrix and computed invariants.
- `orbit_distribution()` – Computes the empirical distribution of Bryant orbits from randomly generated forms.
- `contraction_rank()` – Used extensively to verify the invariance of contraction rank under random $GL(n, \mathbb{R})$ transformations.

Bibliography

[Bry06] Robert L. Bryant, *On the geometry of almost complex 6-manifolds*, Asian Journal of Mathematics **10** (2006), no. 3, 561–605.

[Gla79] E. R. Glass, *A geometric classification of trivectors of seven-dimensional space*, Proceedings of the Royal Society of Edinburgh Section A **83** (1979), no. 1–2, 1–15.

[Gre67] Werner Greub, *Multilinear algebra*, Springer, 1967.

[Har92] Joe Harris, *Algebraic geometry: A first course*, Graduate Texts in Mathematics, vol. 133, Springer-Verlag, 1992.

[Hit00] Nigel Hitchin, *The geometry of three-forms in six and seven dimensions*, Journal of Differential Geometry **55** (2000), no. 3, 547–576.

[HK71] Kenneth M. Hoffman and Ray Kunze, *Linear algebra*, 2 ed., Prentice Hall, 1971.

[KK25] Harish Kishnani and Amit Kulshrestha, *R-weighted graphs and commutators*, 2025.

[Lan12] J. M. Landsberg, *Tensors: Geometry and applications*, American Mathematical Society, 2012.

[Wes81] R. Westwick, *Real trivectors of rank seven*, Linear and Multilinear Algebra **10** (1981), no. 3, 183–204.

[Win10] Sergei Winitzki, *Linear algebra via exterior products*, Lulu Press, 2010.